

Out of Work: Forecasting and Explaining U.S. Unemployment Through Macroeconomic Indicators

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Abstract

Can unemployment be predicted before it rises? This project developed a data pipeline using six public U.S. macroeconomic datasets to forecast unemployment and identify key economic indicators. Results show these indicators are more useful for **early-warning** labor market insight than precise long-term prediction.

Introduction & Motivation

Problem & Question

Economic downturns can quickly raise unemployment, making early warning signals important for planning. This study asks whether U.S. unemployment can be forecast **3, 6, and 12 months** ahead using public macroeconomic indicators, and which indicators **best signal** future labor market changes.

Impact & Stakeholders

The findings can support **policymakers, economists, businesses, investors, and workforce agencies** in making earlier decisions about employment, planning, and resource allocation.

Background & Related Work

- **Macroeconomic indicators** such as GDP, inflation, interest rates, and consumer sentiment are widely used to study unemployment [1].
- **Traditional time-series models** can miss nonlinear relationships, while machine learning can capture more complex economic patterns [2], [3].
- **This study** combines six public U.S. macroeconomic datasets to forecast unemployment and identify early-warning labor market indicators.

Data, Engineering & Ethics

Source & Scale:

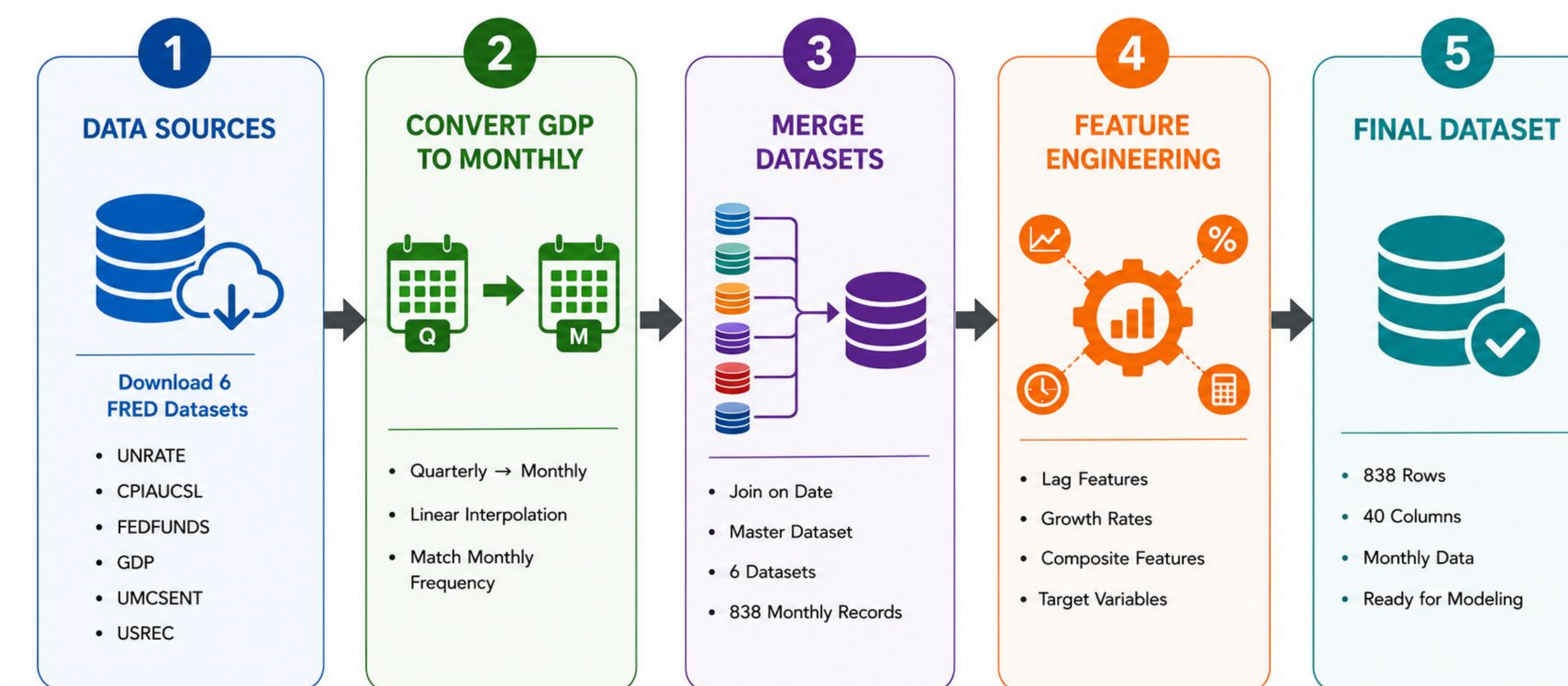
Six public FRED datasets covering **1956–2025** were merged into **838 monthly observations** for forecasting and analysis.

Pipeline:

Download Raw Data → **Clean** → **Merge by Date** → **Feature Engineering** → **Model Dataset**

Ethics:

The project uses only public, aggregated economic data with **no PII**. Results support decision-making but should not replace human judgment.



Methods & Evaluation

Approach:

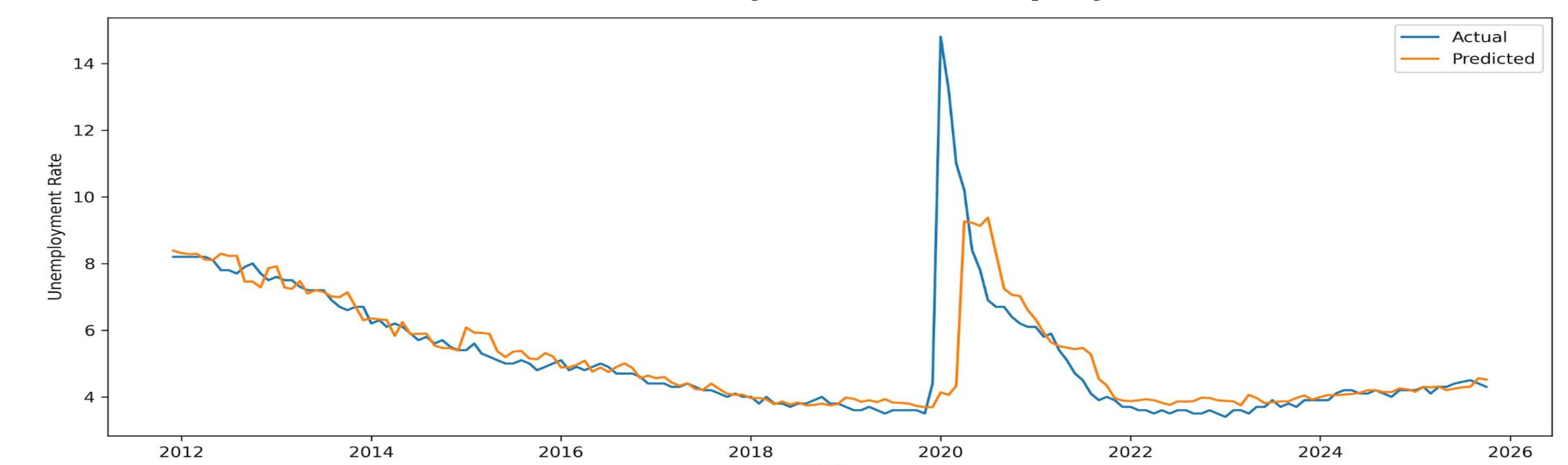
Built an **end-to-end forecasting pipeline** using six macroeconomic datasets, engineered predictive features, and trained **Linear Regression, Random Forest, and Gradient Boosting** models to compare linear and nonlinear forecasting performance.

Validation:

Used a **time-based train/test split** and evaluated results with **MAE, RMSE, and R^2** against an unemployment-only baseline. Feature importance and lead-lag analysis were used to identify the strongest early-warning indicators.

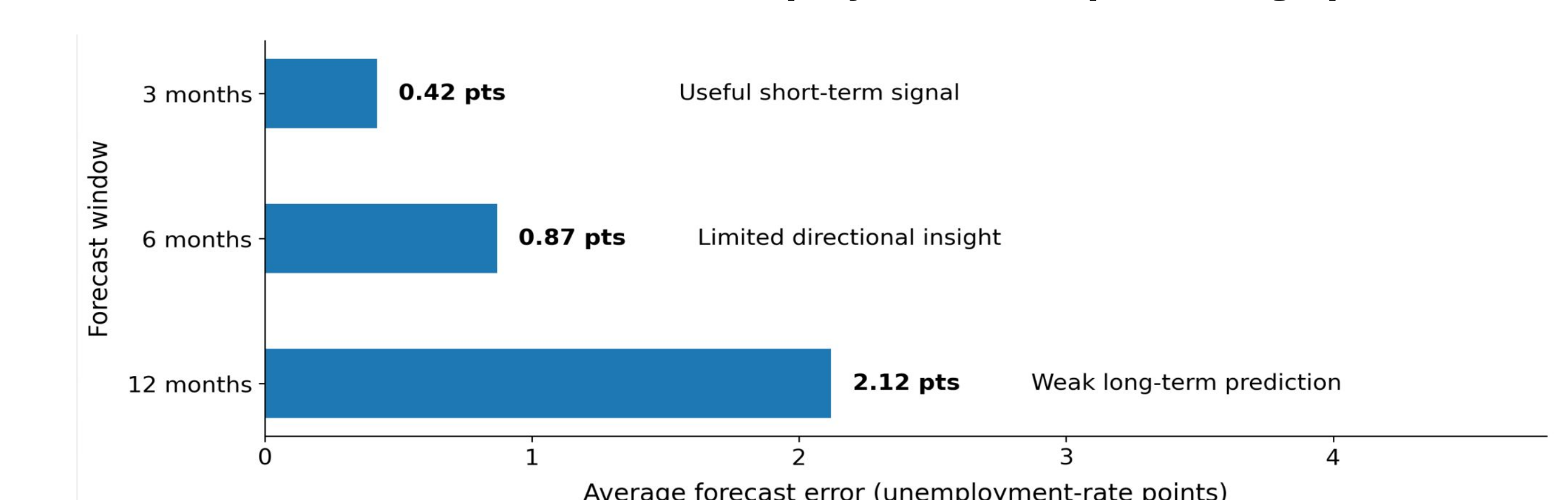
Results & Visualizations

Figure 1. Actual vs. predicted unemployment rate for the 3-month horizon.
Test set: $n = 167$ months; y-axis = unemployment rate (%)



Best 3-month ML model: Gradient Boosting
Average Error (MAE) = 0.42, RMSE = 1.27, Variation Explained (R^2) = 0.51

Figure 2. Average forecast error by 3-, 6-, and 12-month horizons. Errors are MAE values measured in unemployment-rate percentage points



Discussion, Limitations & Conclusions

Discussion:

- The model worked best as a **short-term early-warning tool**, with the 3-month forecast showing useful signal while 6–12 month forecasts became weaker.

Limitations & future work:

- Longer-horizon forecasts were less reliable, so future work should add **state/industry-level labor data** and models better suited for economic shocks.

Conclusions:

- The project answered both research questions: unemployment was most forecastable in the short term, and the strongest historical relationships across business cycles came from **consumer sentiment, recession/economic stress periods, and GDP growth patterns**.

Reproducibility & References

Repository: <https://github.com/manishkallu01-wq/DATA510-Session-2> · Key dependencies: Python 3.13, pandas, NumPy, scikit-learn, matplotlib

References: [1] Federal Reserve Bank of St. Louis, FRED, Online Database, 2026 · [2] Montgomery et al., Time Series Analysis and Forecasting, Wiley, 2015 · [3] James et al., Introduction to Statistical Learning, Springer, 2021.

